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Gradient descent:-

1 of most famous optimization algorithms to perform optimization & by far most common way to optimize neural networks.

→ A way to minimize loss function.

→ Backpropagation algo is basically G.D.

3 types:- accuracy & time (tradeoff).

Batch G.D (Vanilla):-

entire dataset
↓
update.

Let's say 5 epoch;

50 points → Predict.

dot product.

$$\hat{y} = \text{np.dot}(x, w) + b$$

From y & $\hat{y}$ we will find loss

$\left\{ - \sum_{i=1}^{50} \right\} \rightarrow$ on base of this loss
update weight, bias 1 single time.

:- epochs no = no of updates.

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Overall it means if have 5 epochs
update weights & bias 5 times.

Code:- epoch = 10

for i in range(10):

full dataset $\hat{y} = \text{np.dot}(x, w) + b$
here 50 values.

$x = 50$ value.

$\hat{y}, y \rightarrow \text{loss}$.

$\hookrightarrow W, b$ update $W_{\eta} = W_0 - \eta \frac{\partial L}{\partial W}$

$\rightarrow \text{loss}$

total 10 times, w, b updates.

Stochastic G.D:-

epochs $\rightarrow 10$

for i in range(10):

Shuffle.

for i in range($x.\text{shape}[0]$)

$\hookrightarrow$ 1 random point $\rightarrow$ bias

$\hookrightarrow \hat{y} \rightarrow$ forward eliminate.

$\hookrightarrow \text{loss}$

freq of weight update in St. G.D is higher.

→ batch size if $(n) = \text{batch G.D.}$
" " if $(1) = \text{stoch "}$.

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↳ w, b update → $w_n = w_0 - \eta \frac{\partial L}{\partial w}$

print avg loss for each epoch.

$10 \times 50 = 500$ updation.

Note:-

In st. G.D w, b updates → no
of rows time each epoch while
in batch 1 time for 1 epoch.

Differences:-

→ faster:- (given same no of epochs):-

batch as updates are less than st.

faster to converge:- (same epochs):-
move to sol.

Stochastic as takes more updates

so move quickly to sol.

→ loss reduces in st unstably b/c 1 random print decide

→ " " in batch smoothly L.

→ SGD → helps the algo to move out

of local minima. (Don't to be exact

sol } → Negative.

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Vectorization: Without loops using dot

product perform operations $\rightarrow$ Smart replacement
vectorization

$\rightarrow$ faster as it is optimize.

$$np.dot(X, W) + b.$$

$\uparrow$
row

$X \rightarrow$ dataset with 10 crore
 $\downarrow$
didn't load at a time.

Minibatch: Best of both worlds

BGD $\longleftrightarrow$ SGD.

$\swarrow \searrow$
320 rows

in every epoch 10 batches

means 10 update.

batches

$\downarrow$
32

for i in epochs $\rightarrow$

for j in num of batches:

1 batch:

$\hookrightarrow$ X pred (vectorization means

$\rightarrow$ loss

dot product

$\rightarrow$ update

In keras batch size = parameter.

no of batches = no of updates

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faster $BGD > mbgd > SGD$.

converg. $BGD < mbgd < SGD$.

Loss func curve.

Also take vector $\rightarrow$ smaller batches
use

$\rightarrow$ no chance for
stuck.

Why batch size provide in multiple of 2.

2, 4, 8, 32, 128

Ram effective $\rightarrow$ binary cores
So optimization
(faster)